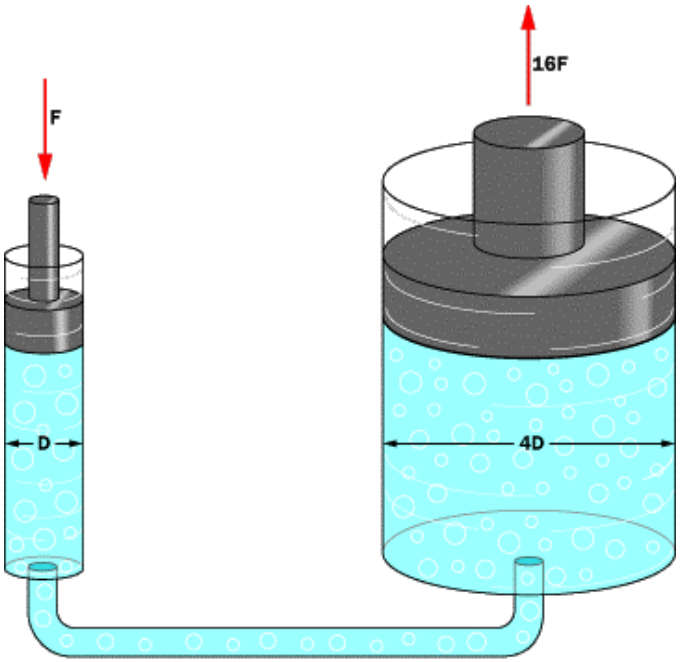
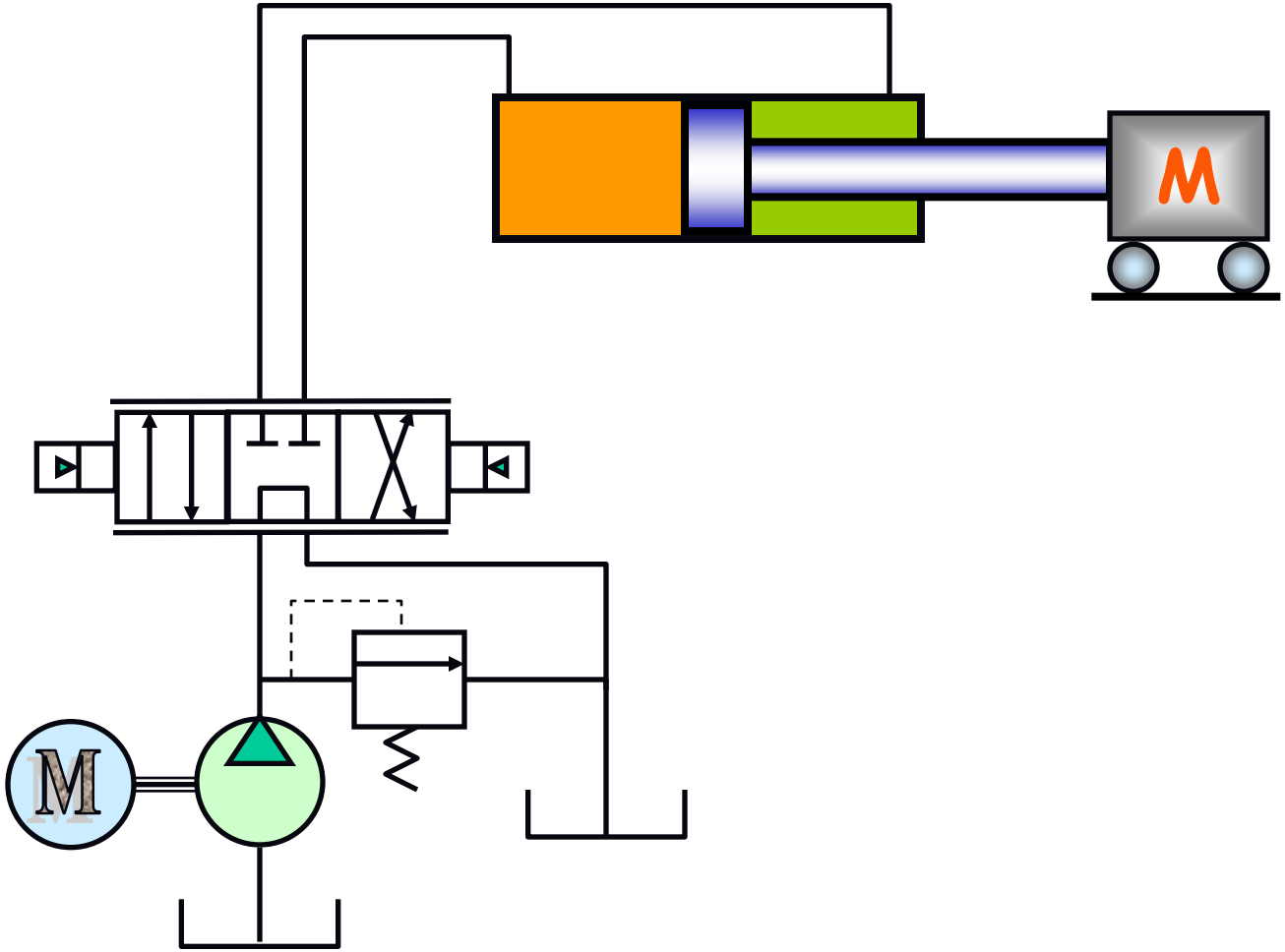
HYDRAULIC SYMBOLS



A Typical Hydraulic Power System



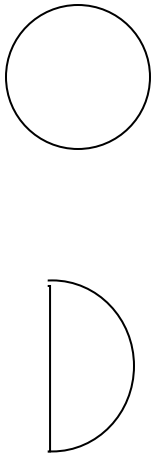
JIC Symbols

- Joint Industry Council

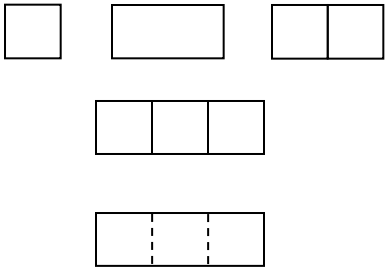
- Symbolic Drawings used in Schematics to Represent Components.

Symbol Groups

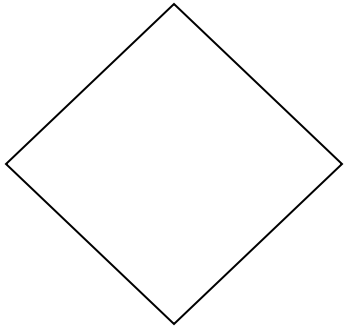
Circles/Semi Circles



Rectangles and Squares



Diamonds

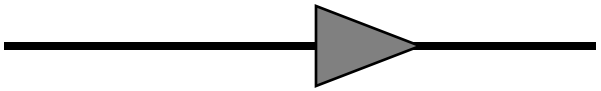


Lines and Connections

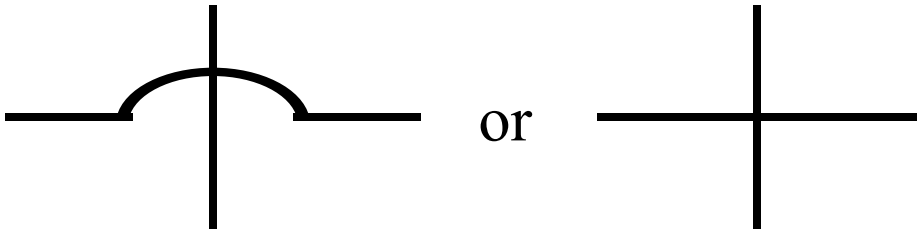
Working Line (Main)

Pilot Control Line

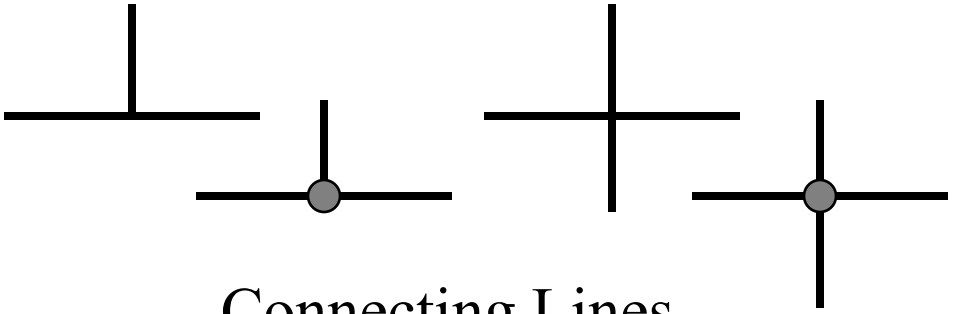
Drain Line



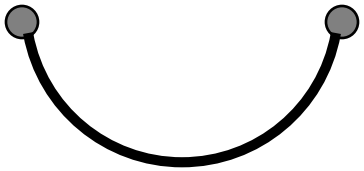
Flow Direction



Crossing Lines



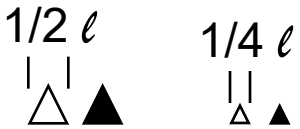
Connecting Lines



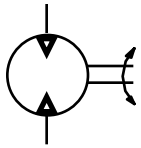
Flexible Line

Functional Elements

Triangle

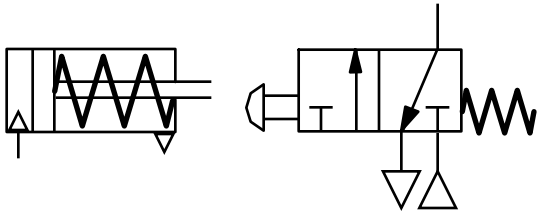


Direction and nature of fluid, open pneumatic or filled hydraulic

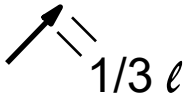


Spring

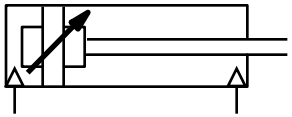
size to suit



Arrow



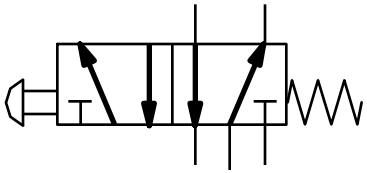
Long sloping indicates adjustability



Functional Elements

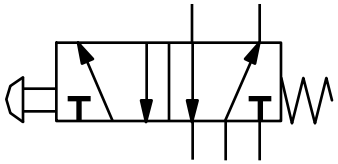
Arrows

Straight or sloping path and flow direction, or motion



Tee T

Closed path or port

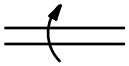


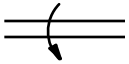
Restriction

Size to suit

Functional Elements

Curved arrows **rotary motion**

Shaft rotation  **clockwise from right hand end**

 **anti-clockwise from right hand end**

both

Seating **90º angle**

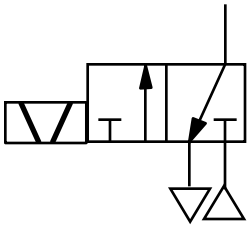
Functional Elements

Temperature

Indication or control size to suit

Operator

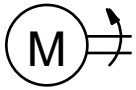
Opposed solenoid windings



Prime mover

M

Electric motor

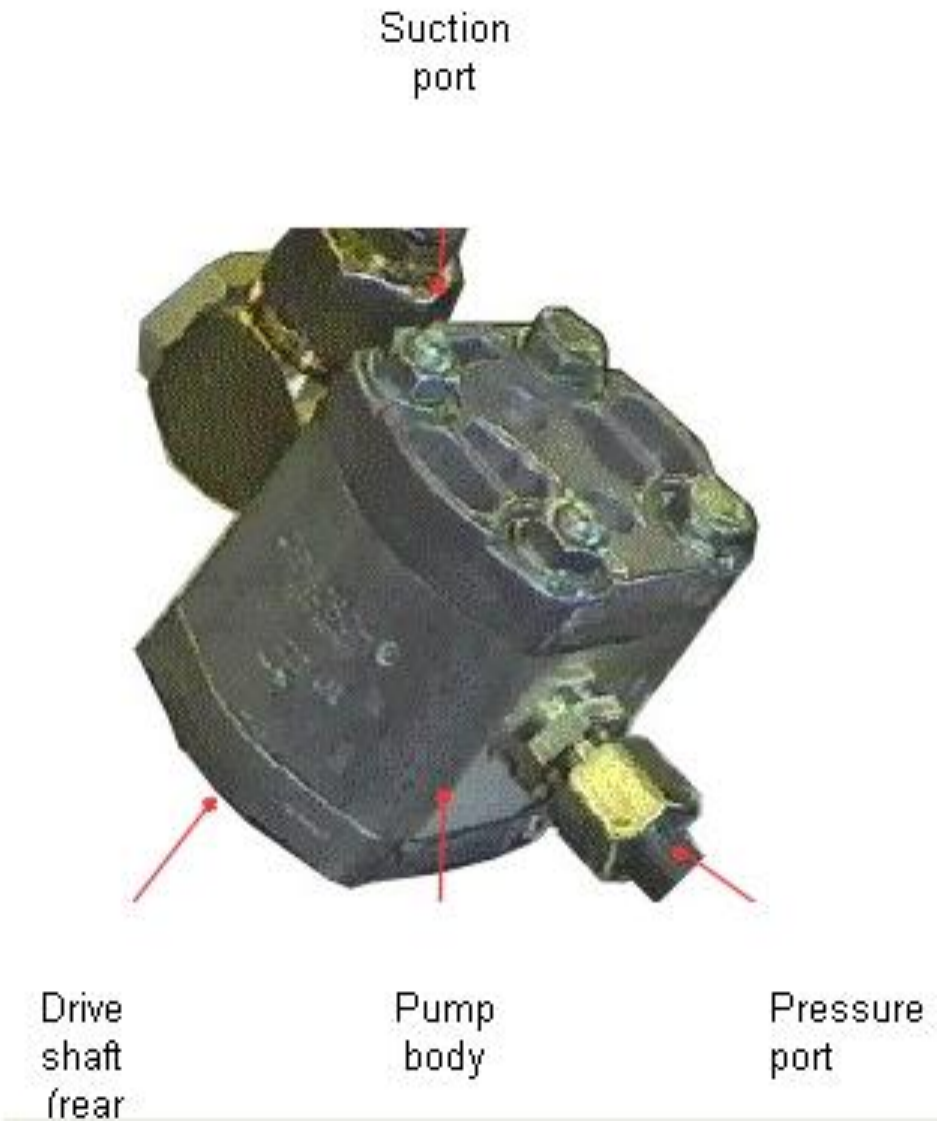


Pumps and Motors

Function

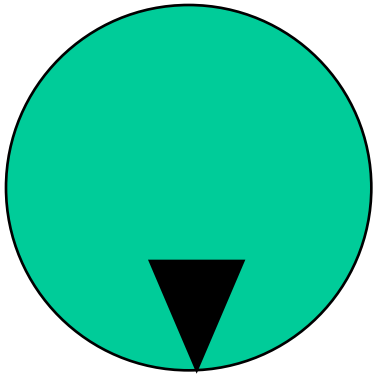
Pump: Conversion of Mechanical to Hydraulic Energy

Hydraulic Motor: Conversion of Hydraulic Energy in to Mechanical Energy of Rotation


Suction
port
Drive
shaft
(rear
Pump
body
Pressure
port

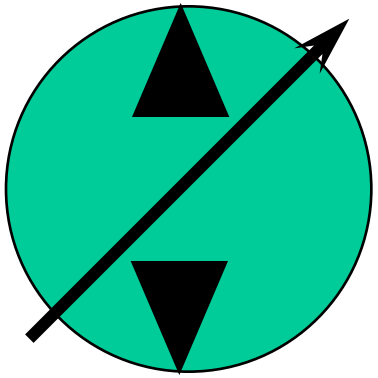
Pumps (JIC Symbols)

- Constant Displacement Single Direction



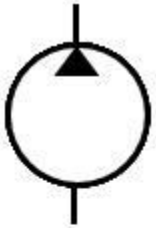
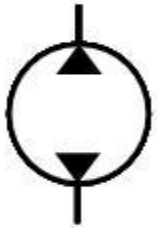
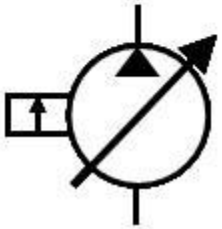
Arrow Showing Oil Flow OUT

- Bi-Directional, Variable Displacement



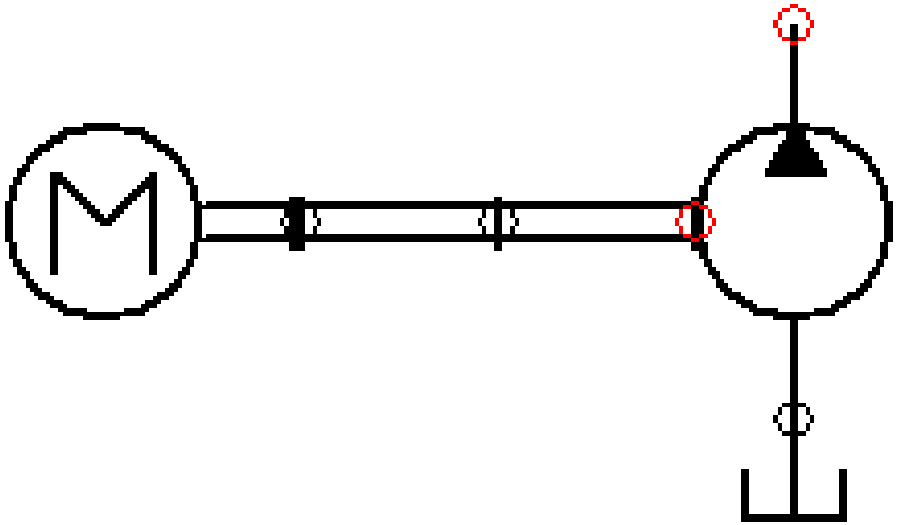
Pumps convert mechanical power into hydraulic force

Fixed & Variable Displacement Pumps

	Fixed displacement hydraulic pump		Variable-displacement hydraulic pump
	Bi-directional fixed-displacement hydraulic pump		Bi-directional Variable-displacement hydraulic pump
			Pressure-compensated Variable-displacement hydraulic pump

Pump Actuation

With Electric Motor



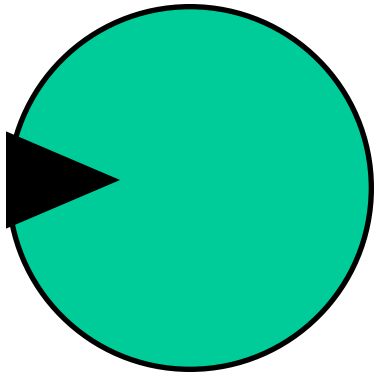
Hydraulic Actuators

Hydraulic Motor: Conversion of Hydraulic Energy into Mechanical Rotations.

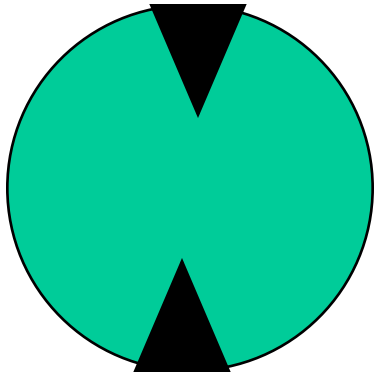
Hydraulic Cylinder: Conversion of Hydraulic Energy in to Mechanical in the Form of Linear Motion

Motors (JIC Symbols)

- Single Direction
- Bi-Directional

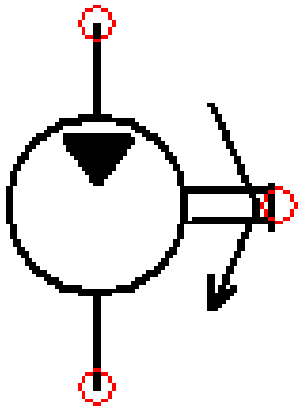


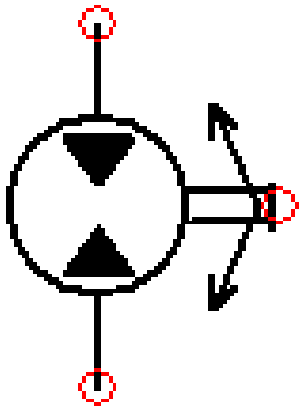
Arrow Showing
Oil Flow IN

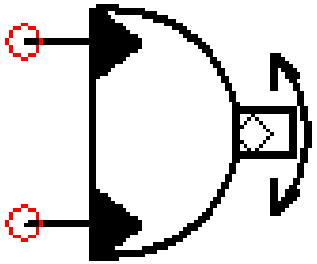


Motors converts hydraulic force into mechanical power

Fixed & Variable Displacement Motors



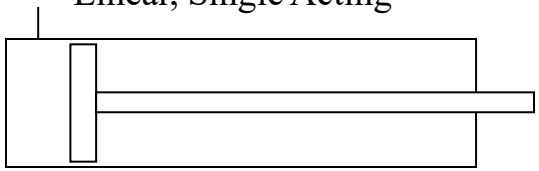




Linear Actuators

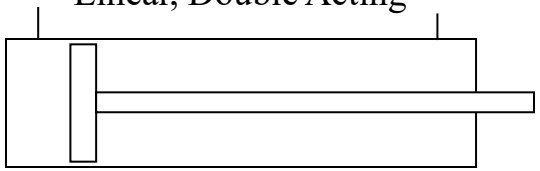
Actuators Can Be:

Linear, Single Acting



Oil pressure extends the cylinder. The load retracts the cylinder.

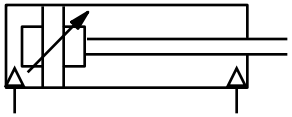
Linear, Double Acting



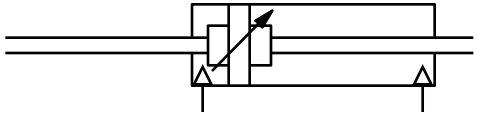
Oil pressure extends the cylinder and retracts the cylinder.

Double acting

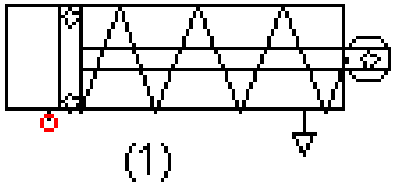
- **Double acting adjustable cushions**



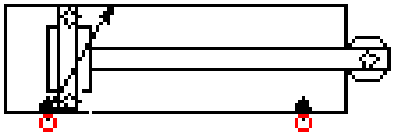
- **Double acting through rod**



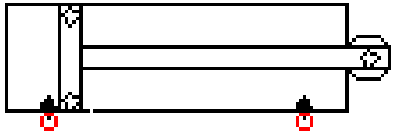
Hydraulic Cylinders



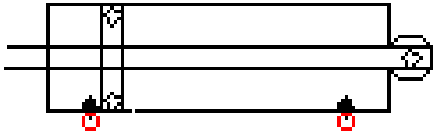
(1)



(2)



(3)



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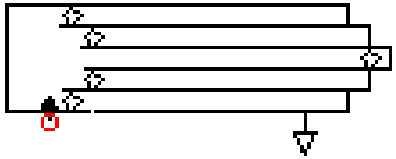
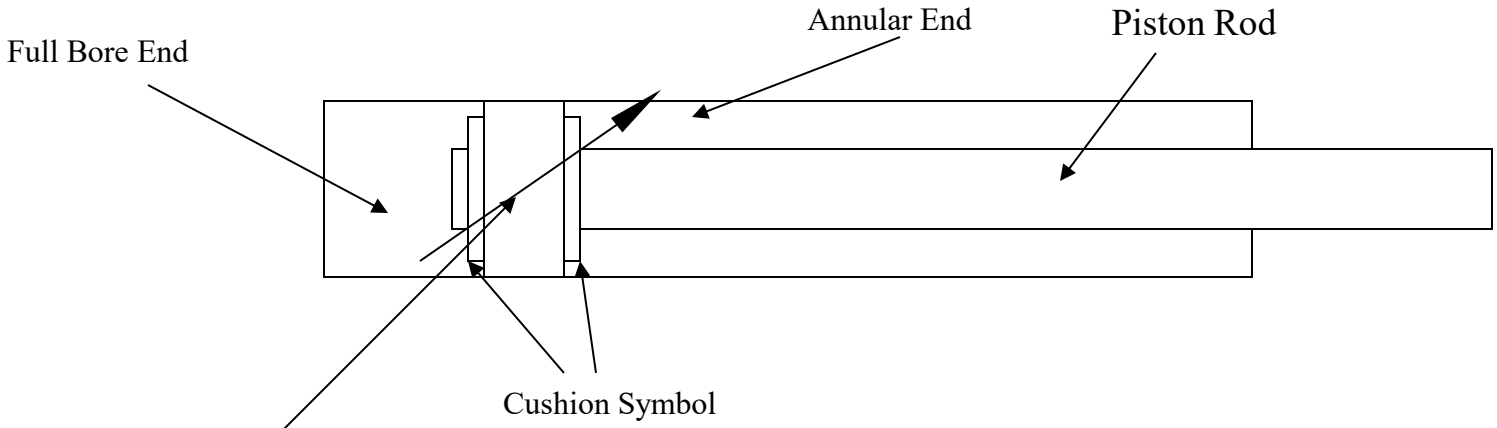
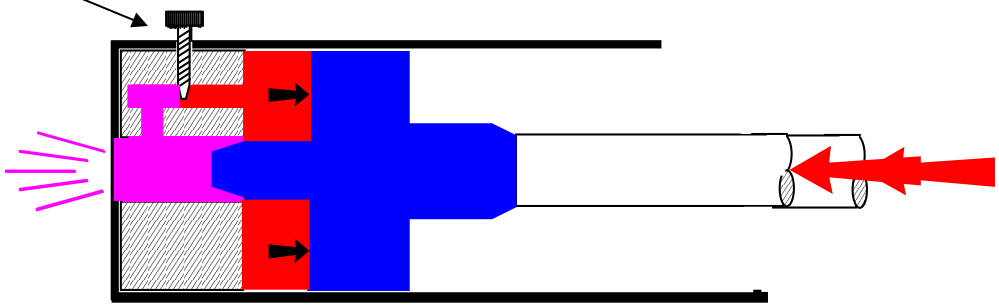


(5)

Cylinder Cushioning



Cushion Illustration

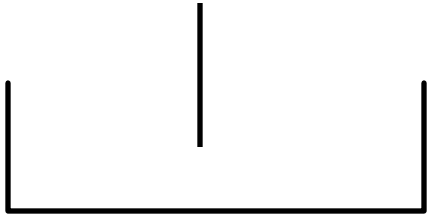


The cushion slows the cylinder down just before it hits the back (or front) of the cylinder.

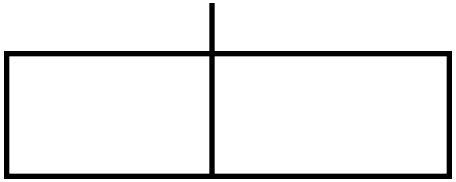
Reservoirs

1. Vented

2. Pressurized



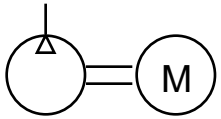
3. Return Above Fluid Level

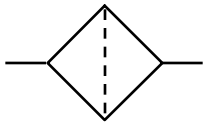


4. Return Below Fluid Level

Pressure Producing Plant

Plant

- Compressor and electric motor 

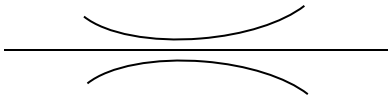
- Air receiver
- Isolating valve
- Air inlet filter 

Other Common Components

Spring

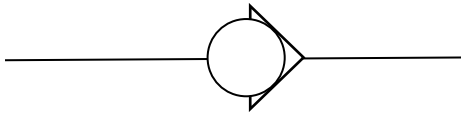
Used to return some valve spools to a known position.

Restrictor



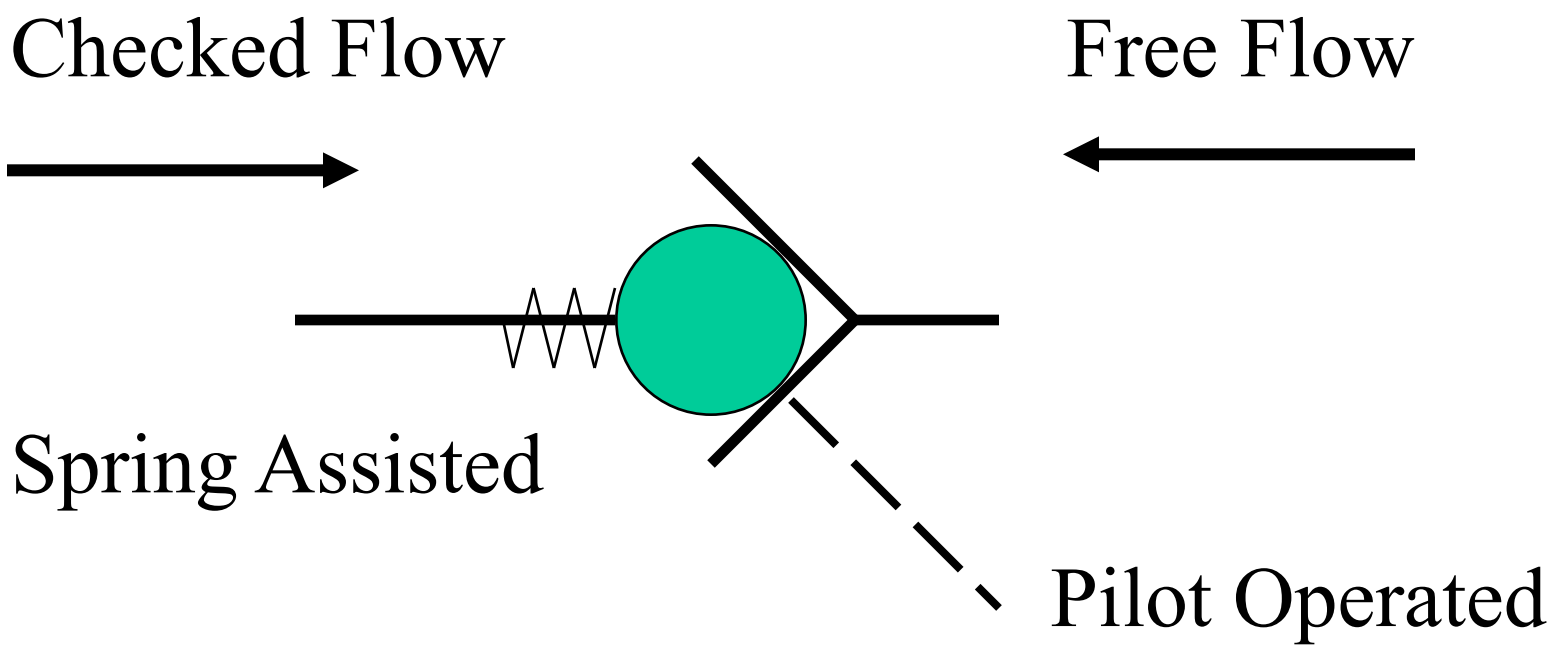
Used to control the flow of oil.

Check Valve

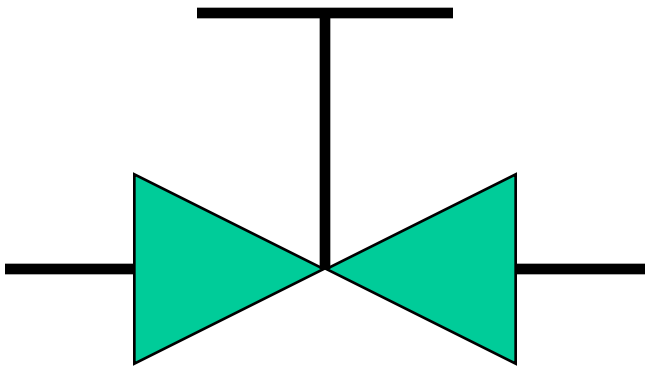


Allows oil in one direction only.

Check Valve



Manual On/Off Valve

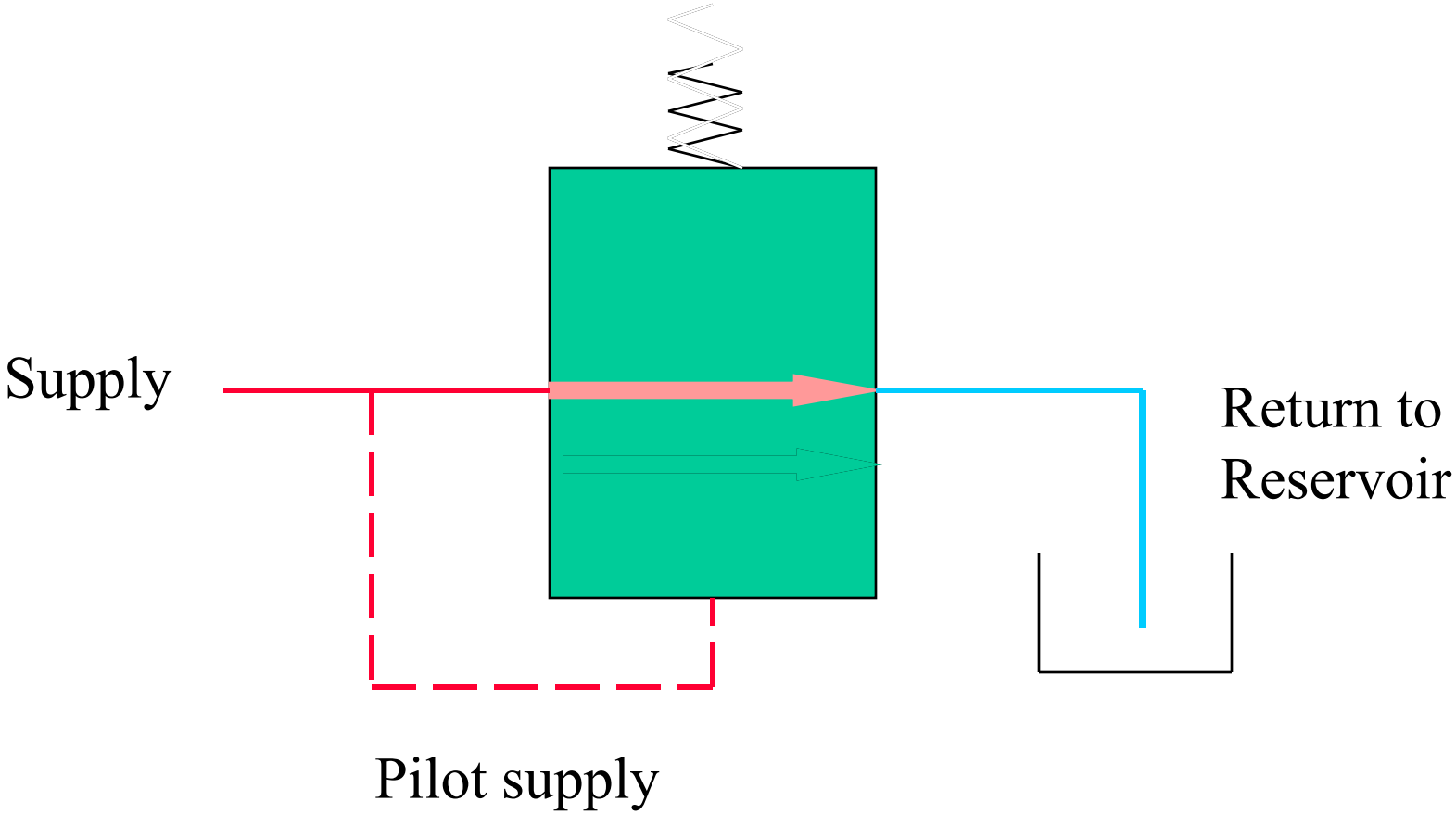


Relief Valves

Protects the Pump and Lines from excessive pressure

Returns fluid back to the reservoir

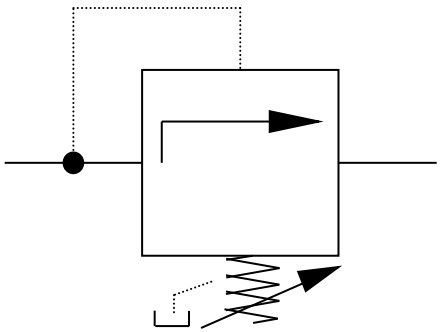
Relief Valve



Control Valves

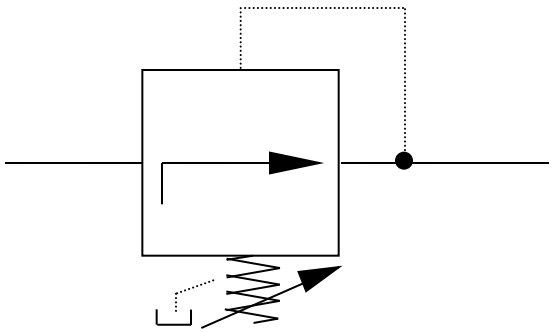
Control Valves Control Pressure and Flow

Relief Valve



Pressure Relief Valves are primarily used to set the maximum system pressure.

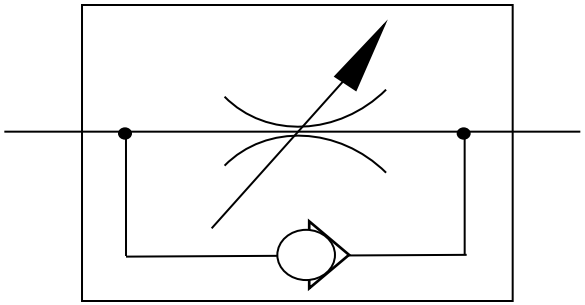
Pressure Reducing Valve



Pressure Reducing Valves are used to set the maximum system pressure at the valve outlet. The valve gives a controlled pressure output lower than the inlet pressure.

Pressure Controlling Valves

Restrictor With Check Valve



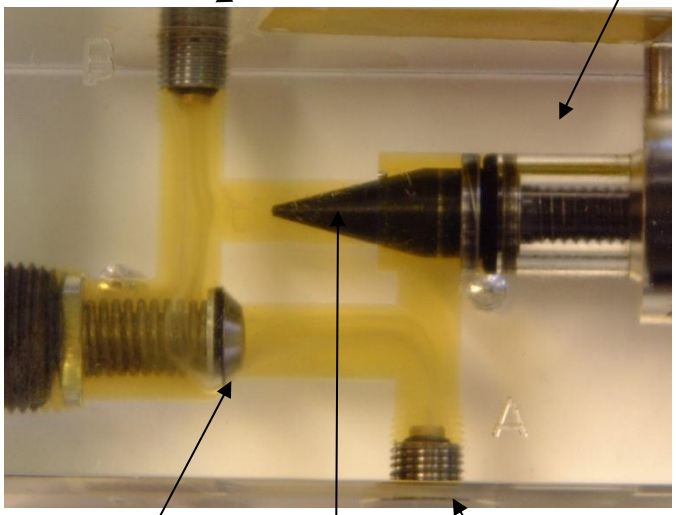
Restrictors are used to control oil flow and thus actuator and system speeds.

Flow Controlling Valve

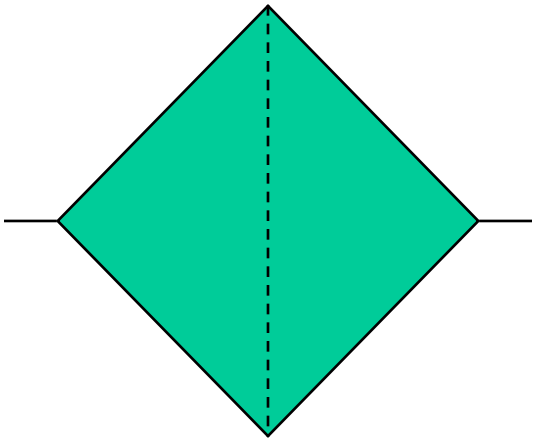
Inside the Relief Valve and Restrictor

Relief Valves

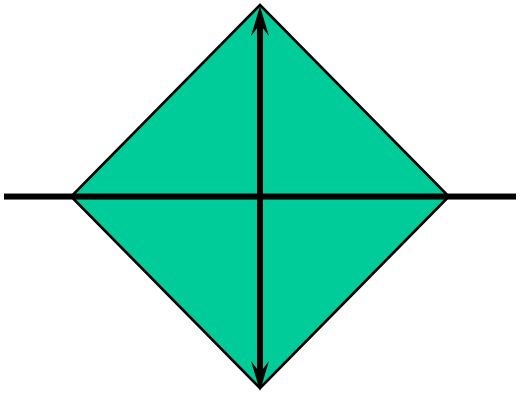
Restrictor with Check Valve



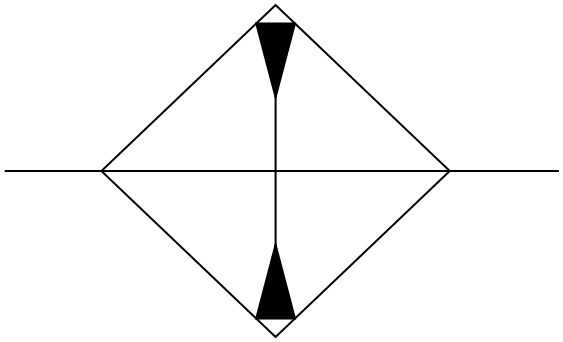
Fluid Conditioners



Filter



Oil Cooler



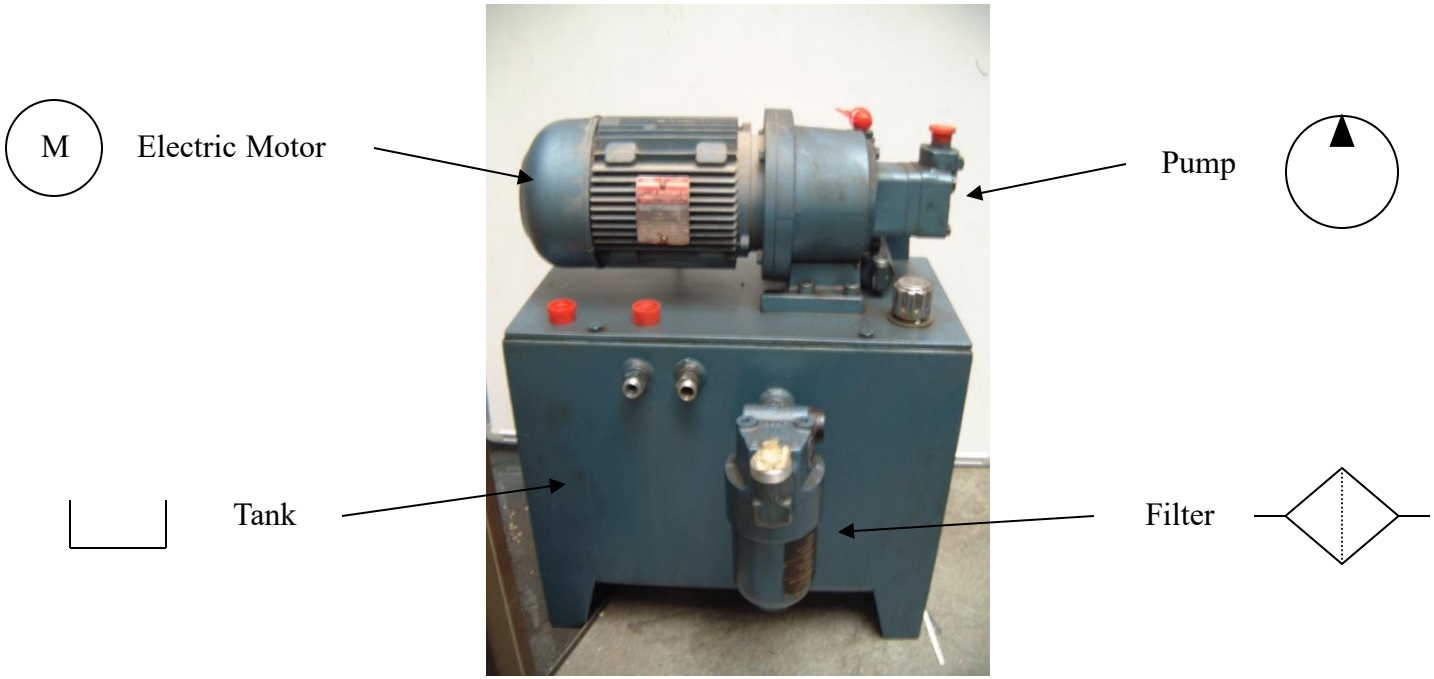
Heater

Gauges

- Oil temperature gauges

- Oil pressure gauges

The Hydraulic Power Pack



Valve symbol structure

Rectangles and Squares

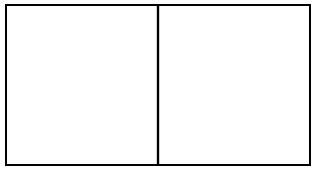
Rectangles and squares are generally used to represent components that direct and control oil in some way.

Valves can be:

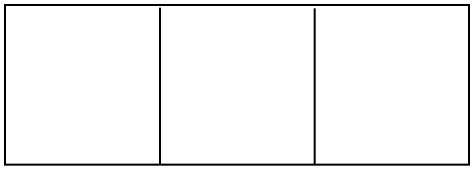
One Position

Usually control. Relieving or controlling pressure and flow for example.

Two Position



Three Position

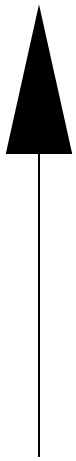


Directing the flow of oil to various positions.

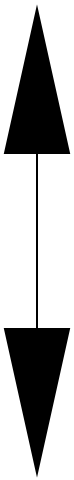
Direction Arrows

Arrows Can Indicate:

Flow in One Direction

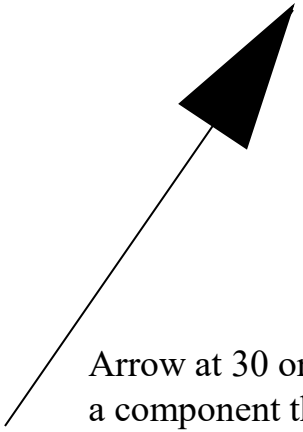


Flow in Two Directions



Rotation

Variability

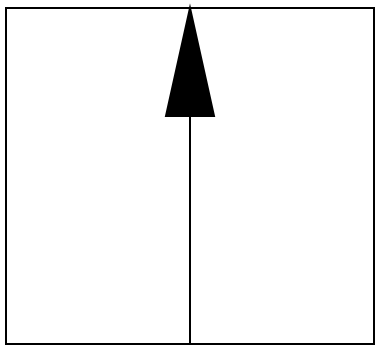


Arrow at 30 or 60 degrees through a component that can be varied.

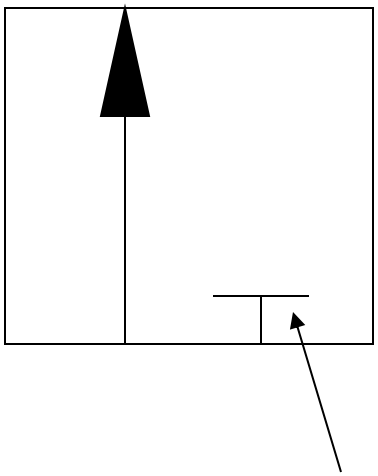
Valves Ports

Valves Can Have Various Ports:

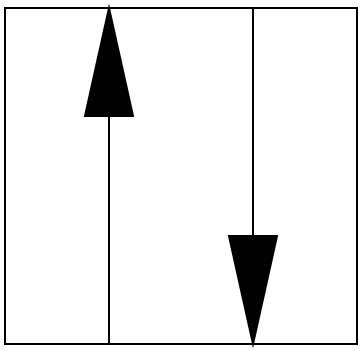
Two Ports



Three Ports



Four Ports



Valve symbol structure

- The function of a valve is given by a pair of numerals separated by a stroke, e.g. 3/2..

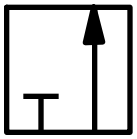
- The first numeral indicates the number of main ports. These are inlets, outlets and exhausts but excludes signal ports and external pilot feeds.

- The second numeral indicates the number of states the valve can achieve.

Valve symbol structure

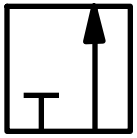
- A 3/2 valve therefore has 3 ports (normally these are inlet, outlet and exhaust) and 2 states (the normal state and the operated state)

- The boxes are two pictures of the same valve

operated  **normal**

Valve symbol structure

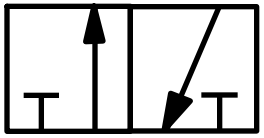
- A valve symbol shows the pictures for each of the valve states joined end to end

operated  **normal**

Valve symbol structure

- A valve symbol shows the pictures for each of the valve states joined end to end

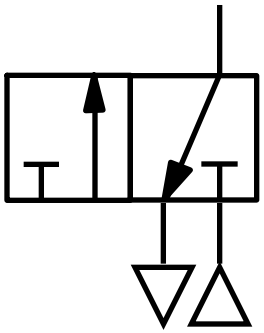
operated



normal

Valve symbol structure

- The port connections are shown to only one of the diagrams to indicate the prevailing state

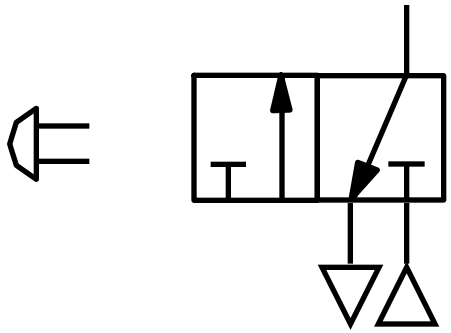


normal

Valve symbol structure

- The operator for a particular state is illustrated against that state

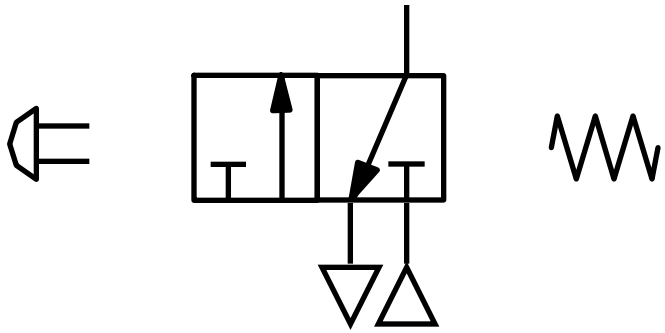
Operated state produced by pushing a button



Valve symbol structure

- The operator for a particular state is illustrated against that state

Operated state produced by pushing a button

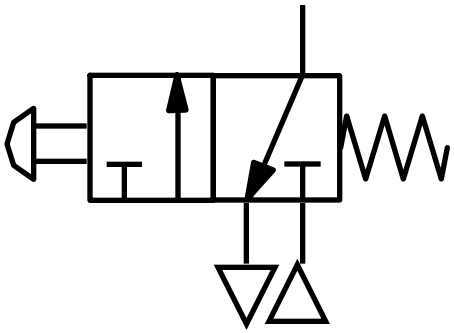


Normal state produced by a spring

Valve symbol structure

- The operator for a particular state is illustrated against that state

Operated state produced by pushing a button



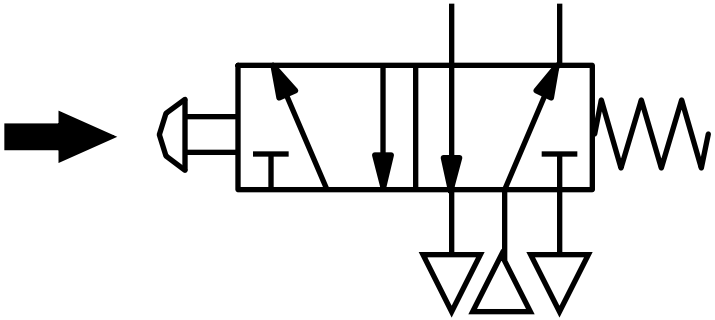
Normal state produced by a spring

Valve symbol structure

- A 5/2 valve symbol is constructed in a similar way. A picture of the valve flow paths for each of the two states is shown by the two boxes. The 5 ports are normally an inlet, 2 outlets and 2 exhausts

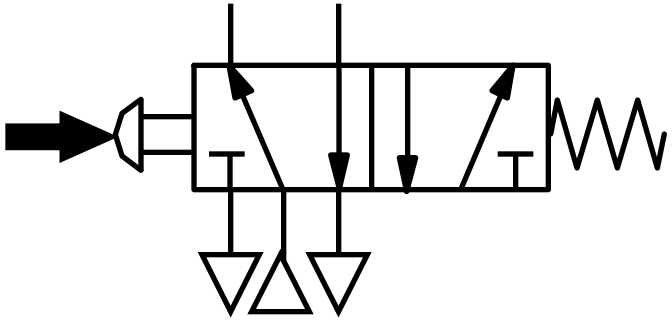
Valve symbol structure

- The full symbol is then made by joining the two boxes and adding operators. The connections are shown against only the prevailing state



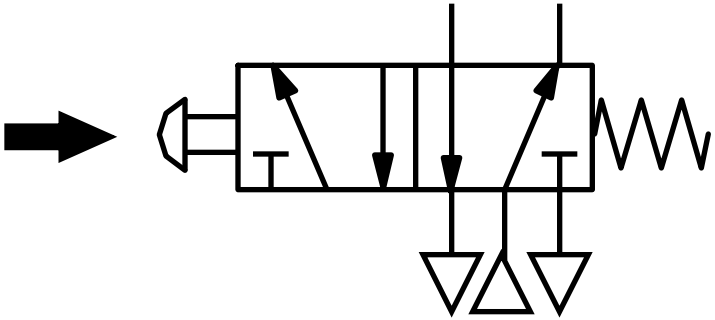
Valve symbol structure

- The full symbol is then made by joining the two boxes and adding operators. The connections are shown against only the prevailing state



Valve symbol structure

- The full symbol is then made by joining the two boxes and adding operators. The connections are shown against only the prevailing state

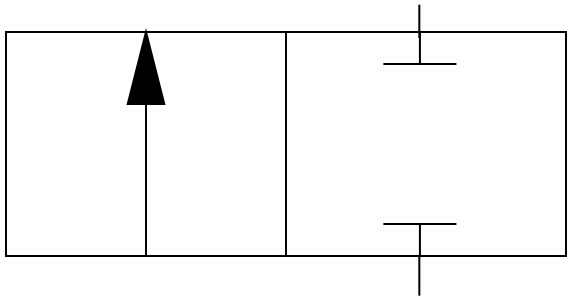


Valve Classification

Valves Can Commonly be Classified as:

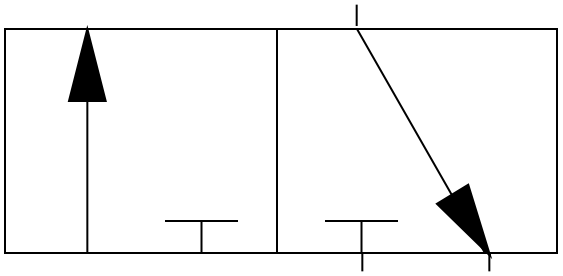
Two Port/Two Position Directional Control Valve (2/2 DCV)

2/2 Wegeventil Sperr Ruhestellung



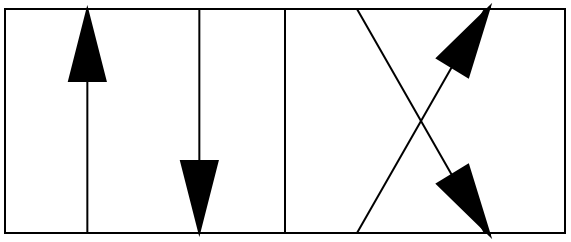
Three Port/Two Position Directional Control Valve (3/2 DCV)

3/2 Wegeventil Sperr Ruhestellung

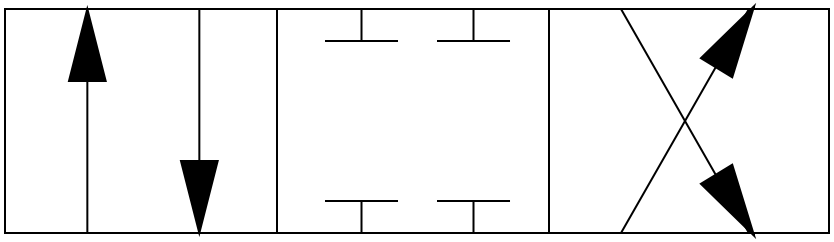


Four Port/Two Position Directional Control Valve (4/2 DCV)

4/2 Wegeventil



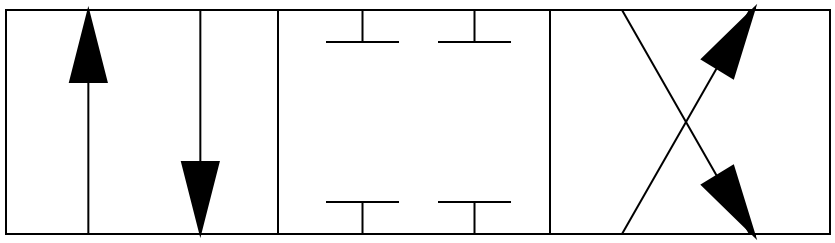
Four Port/Three Position Directional Control Valve (4/3 DCV)



4/3 Wegeventil

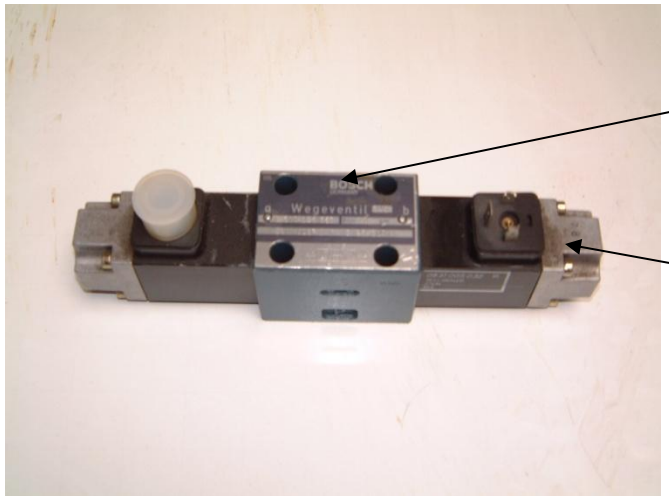
The 4/3 DCV

Symbol



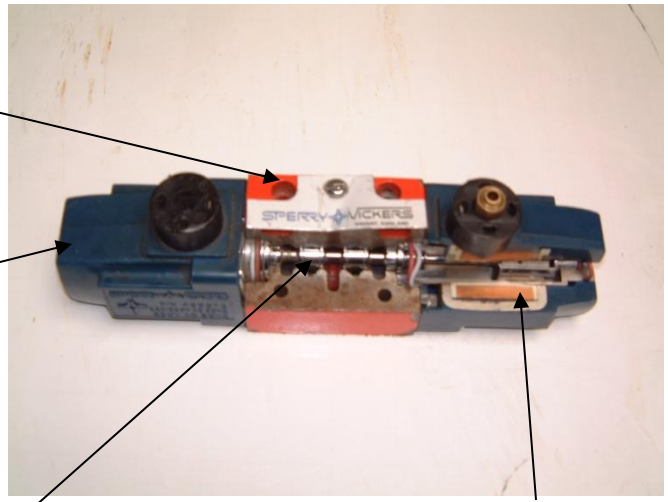
Bosch Solenoid Operated
4/3 DCV

4/3 DCV Cut Away



Valve Body

Solenoid

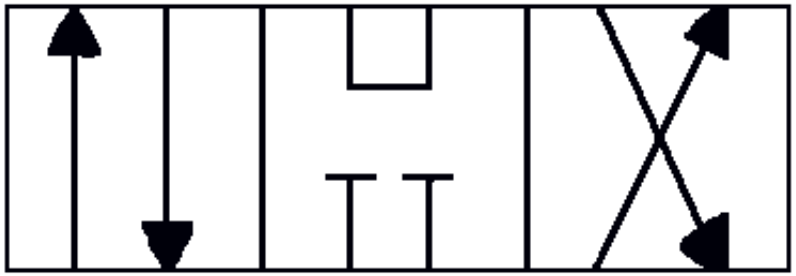


Spool (Schieber)

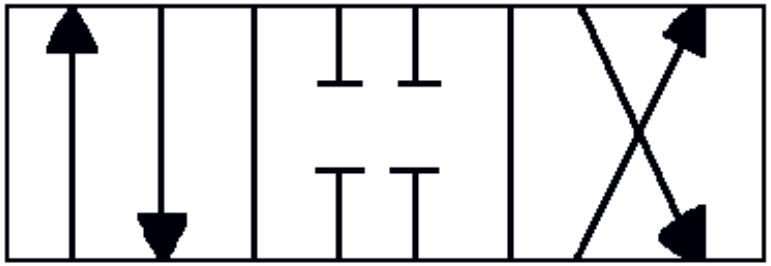
Solenoid
Coil

Control Valves

- Can be tandem (allows flow to tank when closed

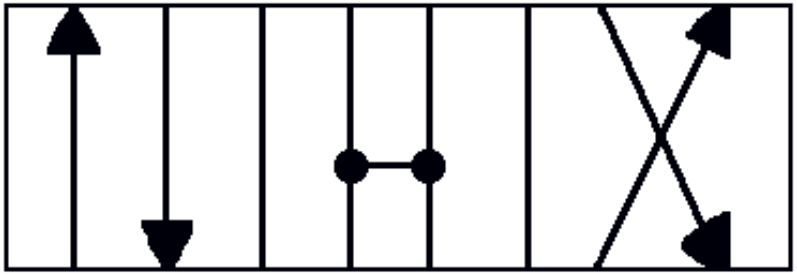


- Can be closed center (allows no flow in the system)



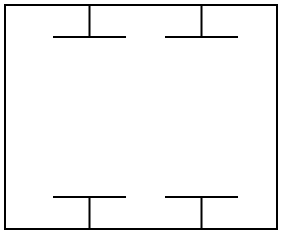
Control Valves

- Can be open center (allows flow on both sides of the valve)

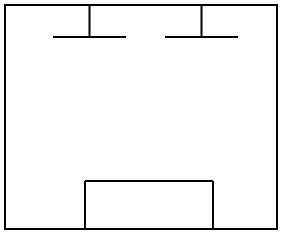


Various 4/3 Centre Configurations

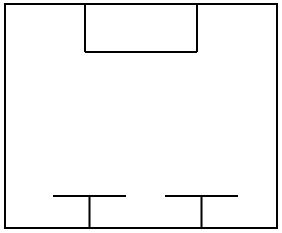
All Blocked



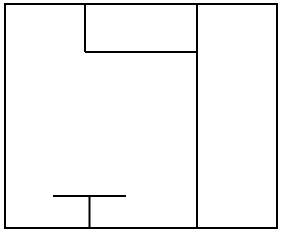
Semi Open Centre
(Pump Side)



Semi Open Centre
System Side)



Y



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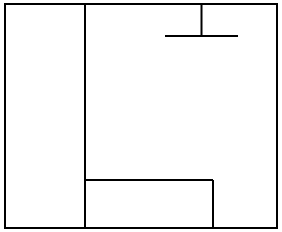

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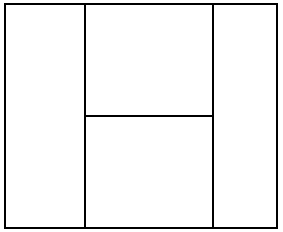
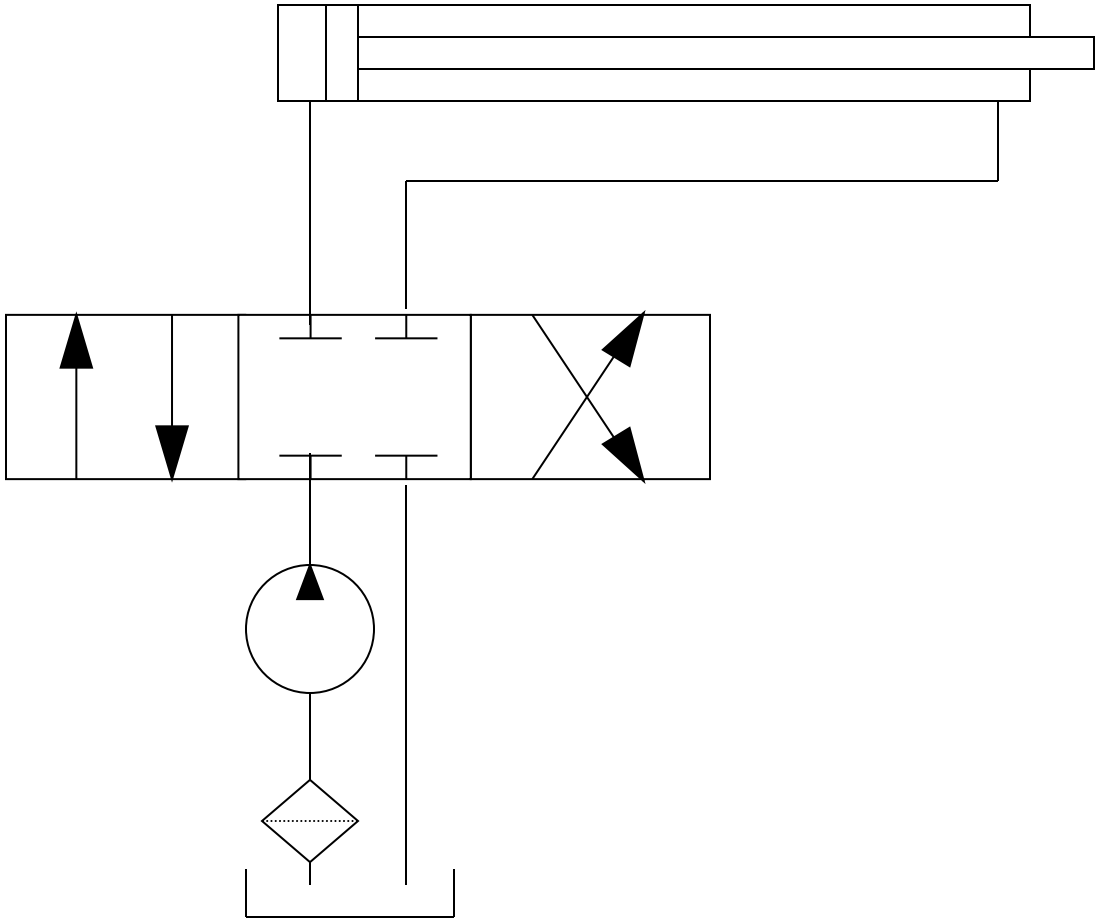


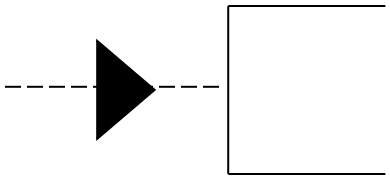
The 4/3 DCV Controlling a Linear Actuator (Cylinder)



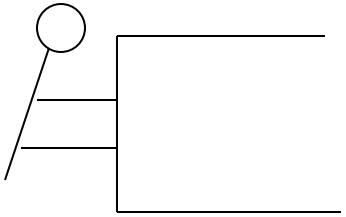
Valve Actuators

The Valve Actuator Moves the Spool to its Various Positions:

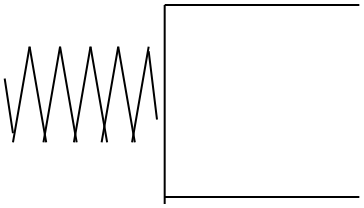
Pilot (Oil Pressure)



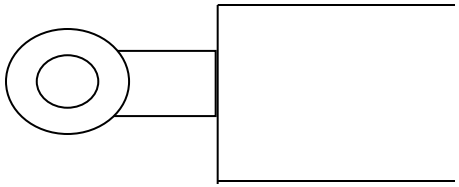
Lever



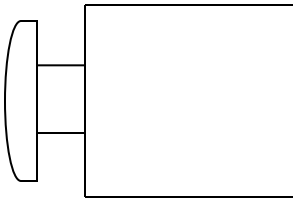
Spring



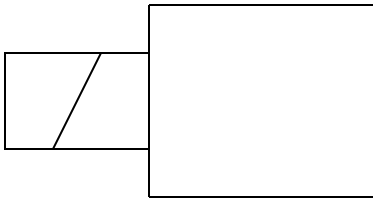
Roller



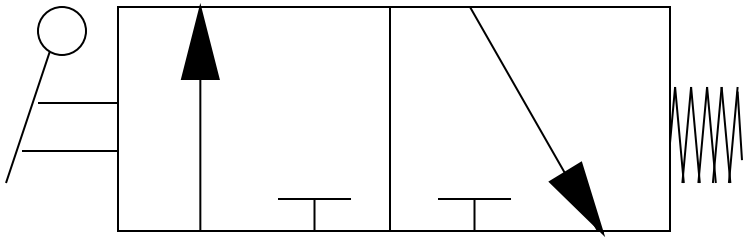
Push Button



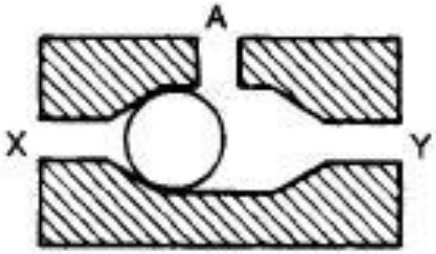
Solenoid



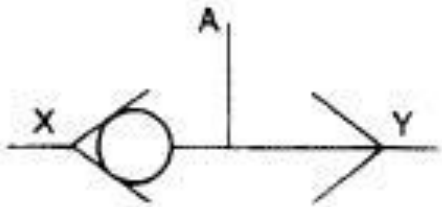
A 3/2 Lever/Spring DCV



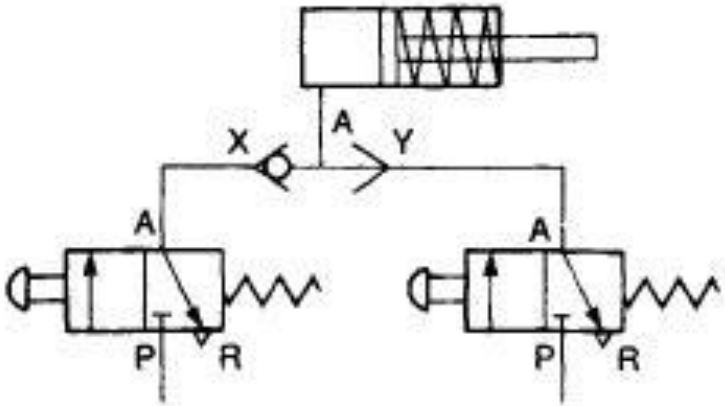
Shuttle Valve



(a) Construction



(b) Symbol



(c) Typical application

Pneumatic shuttle valve

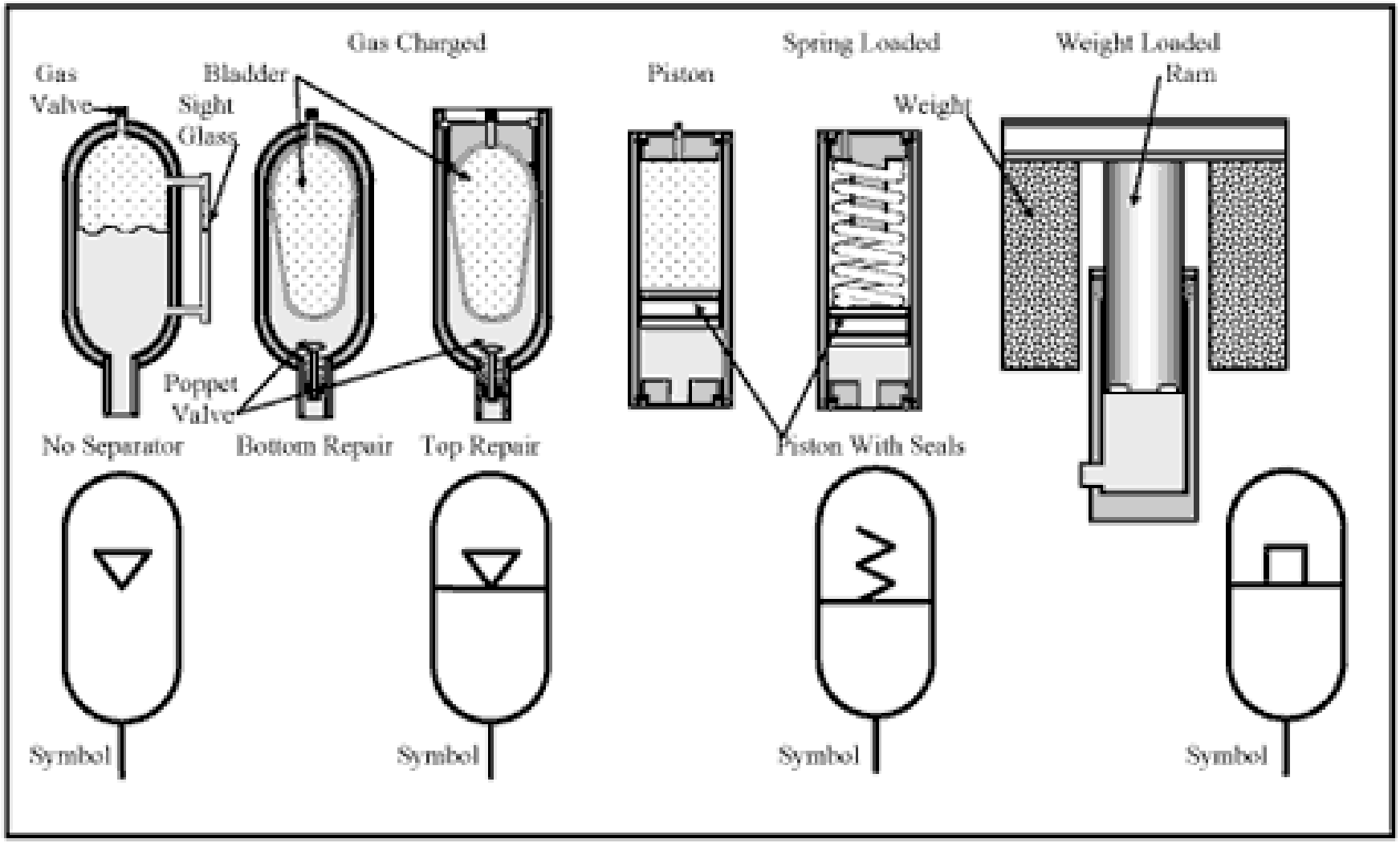
Hydraulic Accumulator

A **hydraulic accumulator** is a pressure storage reservoir in which a non-compressible hydraulic fluid is held under pressure by an external source.

The external source can be a spring, a raised weight, or a compressed gas.

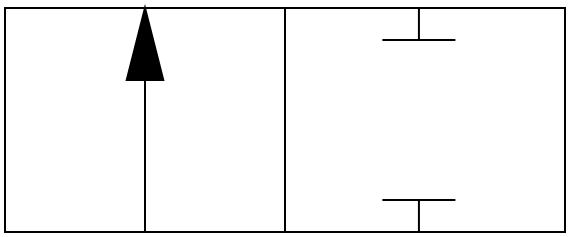
An accumulator enables a hydraulic system to cope with extremes of demand using a less powerful pump, to respond more quickly to a temporary demand, and to smooth out pulsations. It is a type of energy storage device.

Hydraulic Accumulator

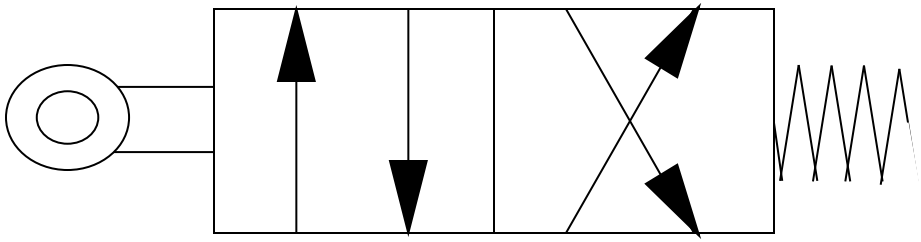


Test

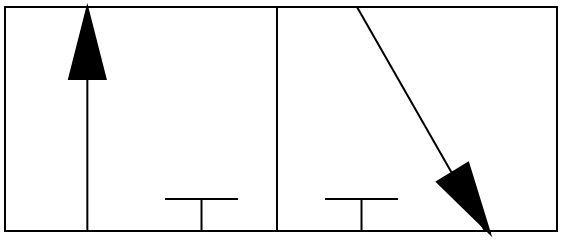
2/2 DCV



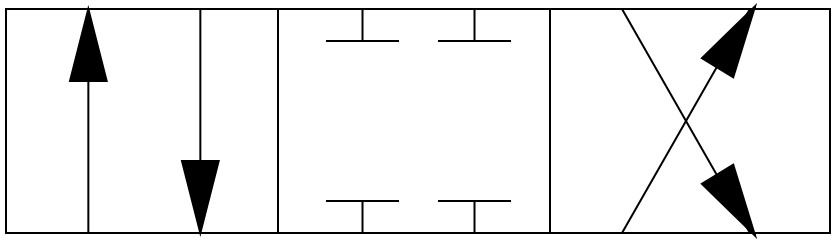
4/2 Roller/Spring DCV



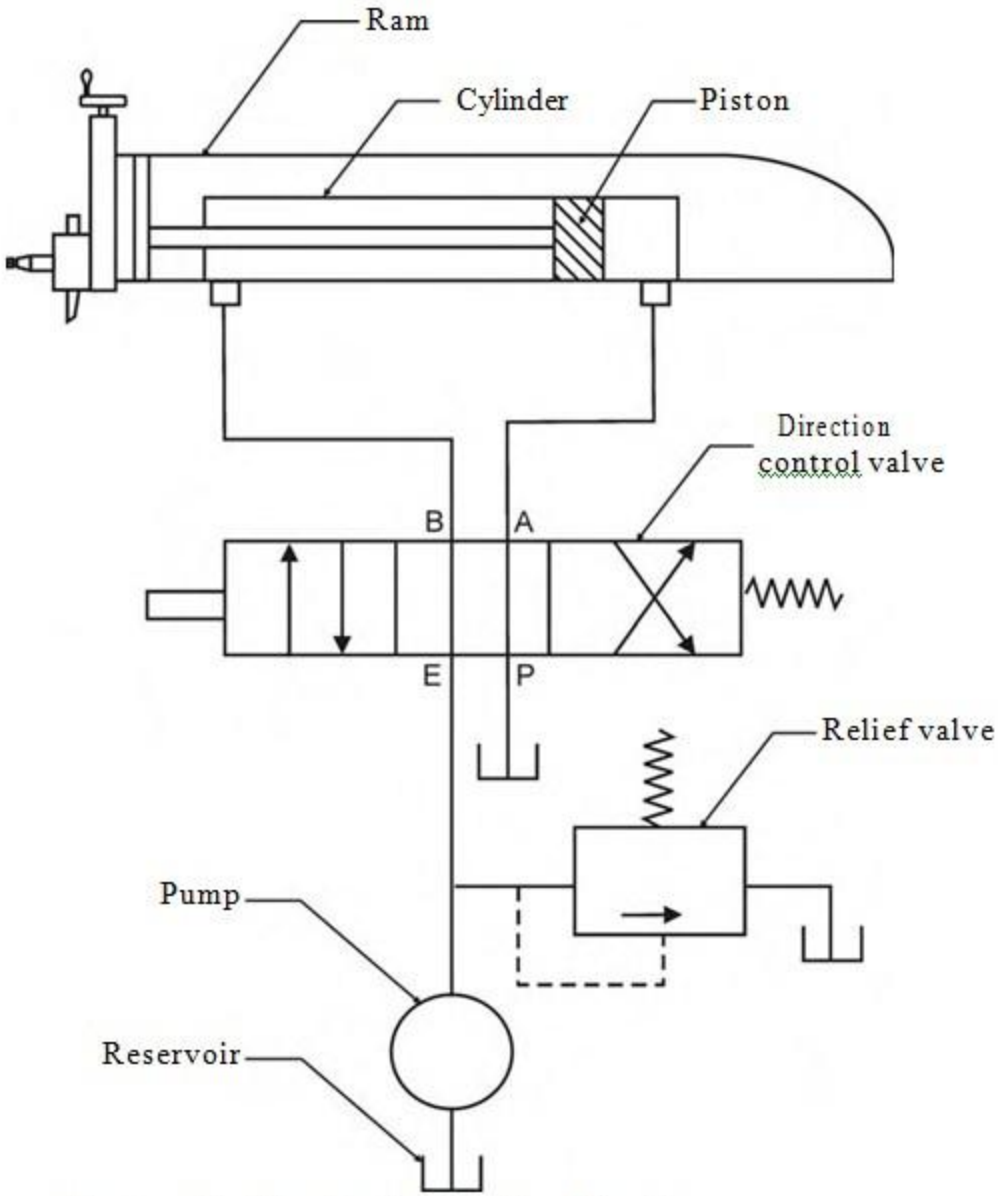
3/2 DCV



4/3 DCV



Hydraulic circuit for a shaper



Hydraulic Circuit of a Shaping Machine

The basic elements in a shaping machine

- Constant discharge hydraulic pump
- Pressure relief valve
- Four way control valve
- Cylinder with piston

- Hydraulic liquid is pumped by a constant discharge pump into the direction control valve. It has four ports and three stages. If the pressure goes beyond desired limits, the same is released by the pressure relief valve.

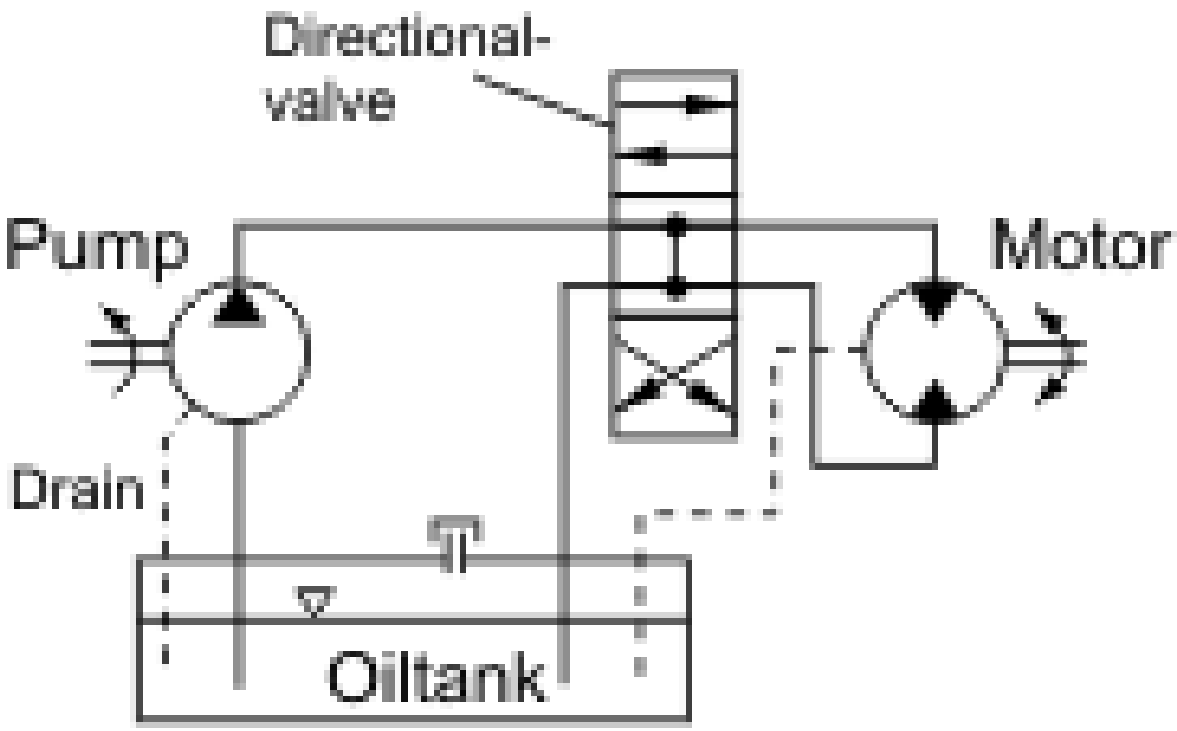
- The liquid reaches the right side of the cylinder through the inlet E and outlet A of the valve. Due to the pressure of the liquid, the piston moves towards left. The toolhead moves forward to perform the forward cutting stroke. The liquid on the left side of the piston is sent back to the reservoir through B and P.

- Due to the change of position of plunger inside the valve, the liquid reaches the left side of the cylinder through the inlet E and outlet B. Piston moves towards right. The toolhead moves in the opposite direction and this movement is known as the return stroke. The liquid on the right side of the piston is sent back to the reservoir through A and P.

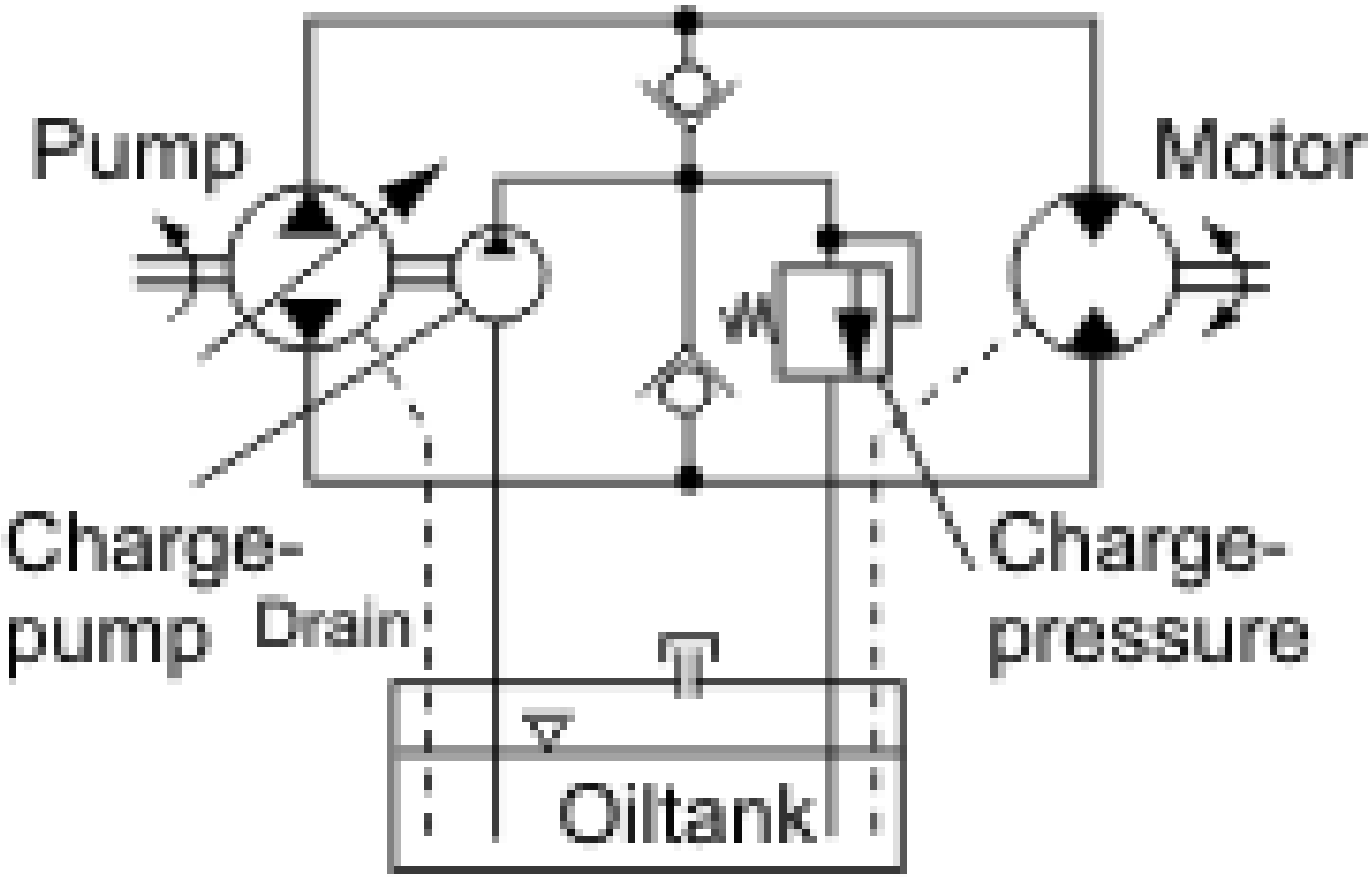
Quick Return of the Ram

- Due to the change of stroke volume of the cylinder during forward and return stroke, the piston returns at a faster rate.
- The stroke volume of the cylinder during return stroke is comparatively smaller than during forward stroke due to the presence of the piston rod. So, the pressure increases automatically to push the piston and the toolhead at a faster speed during return stroke.

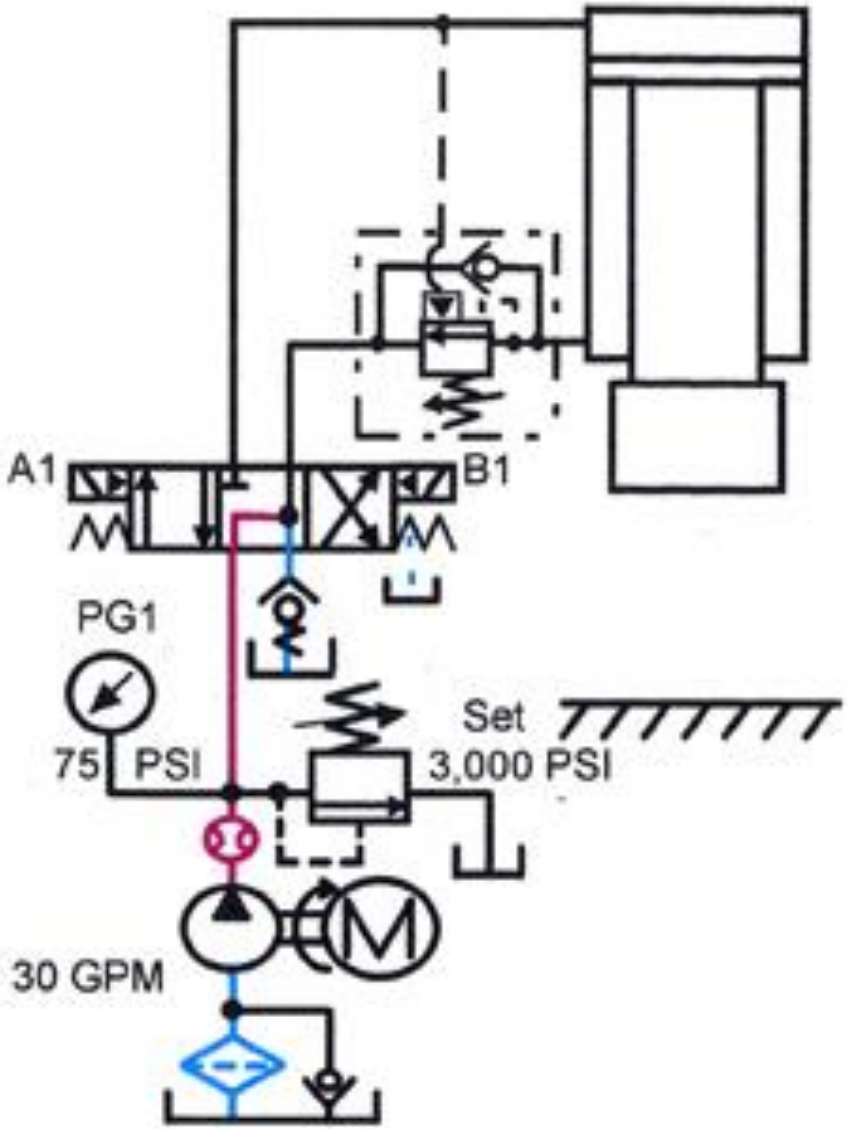
Hydraulic Grinder / Drill

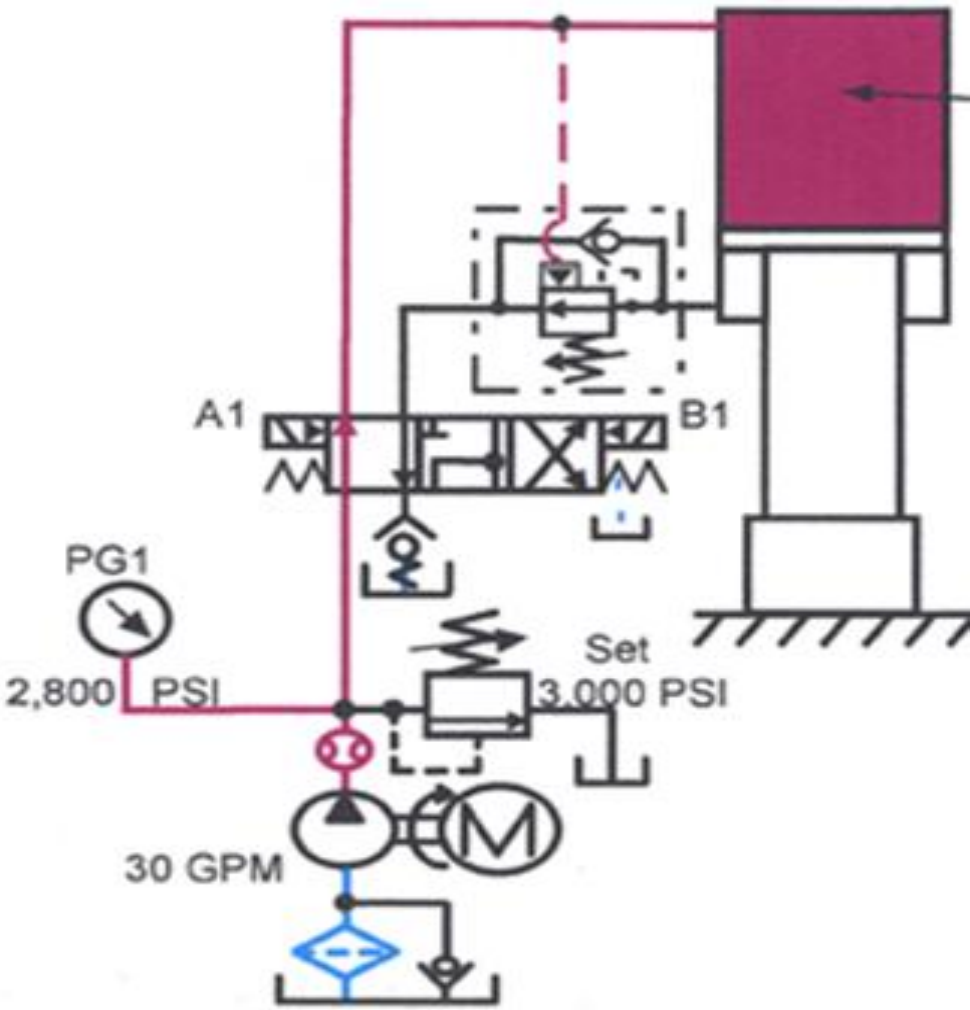


Hydraulic Grinder / Drill



Hydraulic Press





Thank you!